Discrete Fourier Transform (DFT)

complex transform

synthesis

$$x[n] = \sum_{k=0}^{N-1} X[k]\ e^{j2\pi kn/N}$$

analysis

$$X[k] = \frac{1}{N} \sum_{n=0}^{N-1} x[n]\, e^{-j2\pi kn/N}$$

Time domain:
x[n] is complex, discrete and periodic
n runs over one period, from 0 to N-1

Frequency domain:
X[k] is complex, discrete and periodic
k runs over one period, from 0 to N-1
k = 0 to N/2 are positive frequencies
k = N/2 to N-1 are negative frequencies

real transform

synthesis

$$x[n] = \sum_{k=0}^{N/2} ReX[k]\cos(2\pi kn/N) - ImX[k]\sin(2\pi kn/N)$$

analysis

$$ReX[k] = \frac{2}{N} \sum_{n=0}^{N-1} x[n]\cos(2\pi kn/N)$$

$$ImX[k] = \frac{-2}{N} \sum_{n=0}^{N-1} x[n]\sin(2\pi kn/N)$$

Time domain:
x[n] is real, discrete and periodic
n runs over one period, from 0 to N-1

Frequency domain:
Re X[k] is real, discrete and periodic
Im X[k] is real, discrete and periodic
k runs over one-half period, from 0 to N/2

Note: Before using the synthesis equation, the values for Re X[0] and Re X[N/2] must be divided by two.

Discrete Time Fourier Transform (DTFT)

complex transform

synthesis

$$x[n] = \int_0^{2\pi} X(\omega)\, e^{j\omega n}\, d\omega$$

analysis

$$X(\omega) = \frac{1}{2\pi} \sum_{n=-\infty}^{+\infty} x[n]\, e^{-j\omega n}$$

Time domain:
x[n] is complex, discrete and aperiodic
n runs from negative to positive infinity

Frequency domain:
X(ω) is complex, continuous, and periodic
ω runs over a single period, from 0 to 2π
ω = 0 to π are positive frequencies
ω = π to 2π are negative frequencies

real transform

synthesis

$$x[n] = \int_0^{\pi} ReX(\omega)\cos(\omega n) - ImX(\omega)\sin(\omega n)\, d\omega$$

analysis

$$ReX(\omega) = \frac{1}{\pi} \sum_{n=-\infty}^{+\infty} x[n]\cos(\omega n)$$

$$ImX(\omega) = \frac{-1}{\pi} \sum_{n=-\infty}^{+\infty} x[n]\sin(\omega n)$$

Time domain:
x[n] is real, discrete and aperiodic
n runs from negative to positive infinity

Frequency domain:
Re X(ω) is real, continuous and periodic
Im X(ω) is real, continuous and periodic
ω runs over one-half period, from 0 to π

TABLE 31-1 The Fourier Transforms